Higher National Diploma in Engineering (Electrical and Electronics)

First Year, First Semester Examination — 2018

Electrical Principles B - 2018

Question 1

(a) Define the following terms related to a network graph: (i) **Directed Graph** (ii) **Tree** (iii) **Co-tree** (iv) **Twig**. [2 marks]

(b) For the circuit shown in Fig. Q1(a): (i) Draw the **directed graph** of the network. [2 marks] (ii) Draw a **tree** by selecting branch no: 2 and no: 4 from the oriented graph. [1 mark] (iii) Write the **cut set matrix**. [2 marks] (iv) Write the **branch admittance matrix**, $[Y]$. [2 marks] (v) Write the **branch current source matrix**. [1 mark] (vi) Write the **branch voltage source matrix**. [1 mark] (vii) Write the **twig voltage matrix**. [1 mark] (viii) Set the **node pair equations** using the formula: $[Q][Y][Q]^T[V] = [Q][i_s] - [Q][Y_b][v_s]$. Find the value for v. [2 marks]

(c) For the circuit given in Fig. Q1(b), obtain the **node voltages** V_A, V_B **and** V_C by using nodal equations. (Resistor values are in ohms). [6 marks]

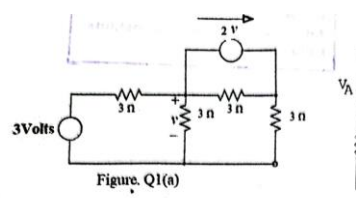

Figure. Q1(a)

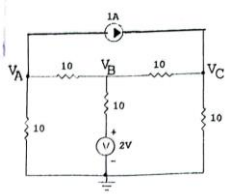

Figure. Q1(b)

Question 2

(i) State the **construction difference** between single transient circuit and double transient circuit. [2 marks] (ii) List down the type of **surges** occurred in the overhead transmission lines and briefly discuss the causes for generation of those surges. [4 marks] (iii) Draw the **lightning surge wave form** relevant to the data given below: $10\mathrm{kA}, 2.5\mu\mathrm{sec}, 25\mu\mathrm{sec}$. [3 marks] (iv) A $50\mathrm{Hz}$ sinusoidal voltage of maximum value $100\mathrm{volts}$ is applied to a series circuit of resistance 5Ω and inductance of $0.2\mathrm{H}$. Find an **expression for the value of the current at any instant** after the voltage is applied. Assuming that voltage is zero at the instant of application. Calculate the **current value** 0.05 **second after switch on**. [6 marks] (v) The R-L-C series circuit contains a coil of inductance $L = 423\mathrm{mH}$, resistance $R = 80\Omega$ and capacitance $C = 169.2\mu\mathrm{F}$. A step voltage of $2\ \mathrm{V}$ is applied to the circuit. Assume that current $i = 0$ at the instant of applying the step voltage. Then determine: (a). the **state of damping** (b). an **expression for current** i (c). Is it possible to obtain **critical damping** for above circuit? If so, how do you obtain critical damping?

Question 3

(a) (i) The critical frequency of a **low pass active filter** is f_c. Mark the **Bandwidth** of the filter on filter response curve. [2 marks] (ii) What is the benefit of **increasing the roll off rate** of active low pass filter? [1 mark] (b) (i) Design a **high pass passive filter** having a cut-off frequency of $2\ \mathrm{kHz}$ with a load resistance of 300Ω in: (i) T - configuration (ii) π - configuration. [4 marks] (ii) An active low pass filter network having a roll off rate of $-20\mathrm{dB/decade/pole}$. Draw the suitable **active filter network** in which the roll off rate increase to $-60\mathrm{dB/decade/pole}$. [3 marks] (iii) Determine the **nominal impedance** and the **cut off frequency** of filter circuits given in Figure Q3(a) and Figure Q3(b) below. [10 marks]

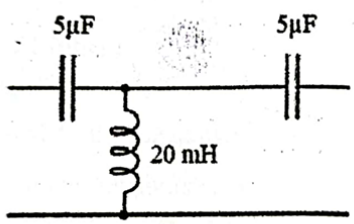

Figure.Q3(a)

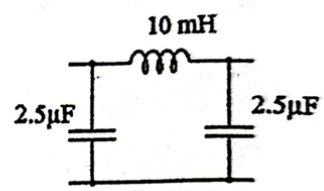

Figure.Q3(b)

Question 4

(a) (i) Write the **voltage equation** at any distance of the mismatched transmission line. [1 mark] (ii) What is the **correctly terminated transmission line**? [1 mark] (iii) Define the following terms related to the mismatched transmission line: (I) **Reflection coefficient**, (II) **Standing wave ratio (SWR)**. [3 marks] (iv) A transmission line has a characteristic impedance of 10Ω and negligible loss. If the terminating impedance of the line is $(6+j4)\Omega$, determine: (I) the **reflection coefficient**, (II) the **standing wave ratio (SWR)**. [3 marks] (b) (i) Draw the **two equivalent circuits** when $\lambda/4$ line fulfill the following conditions: (I). Length less than $\lambda/4$ terminating end is short circuit and other end is open circuit. (II). Length less than $\lambda/4$ terminating end is open circuit and other end is also open circuit. [2 marks] (ii) State the **demerit of single stub machine** in transmission line. [2 marks] (iii) Briefly explain the method to be adopted to **overcome the disadvantage of single stub machine**. [2 marks] (iv) A cable has the following primary constants: Resistance, $R = 80\Omega/\mathrm{loop\ km}$; Conductance, $G = 2\mu\mathrm{S/loop\ km}$; Capacitance, $C = 5\ \mathrm{nF/loop\ km}$. Determine the following for **minimum distortion** at a frequency of $1.0\ \mathrm{kHz}$: (I) the value of **inductance per loop km** required (II) **Characteristic Impedance**, Z_0 (III) **Propagation coefficient**, γ (IV) **attenuation coefficient**, α and **Phase delay**, β (V) **Wavelength**, λ on the line (VI) **velocity of propagation of signal**. [6 marks]

Question 5

(a) (i) List out the **two types of optical sources** commonly used in the fiber optic communication and briefly describe the materials used for their construction. [4 marks] (ii) Briefly explain the **process of data transmission** through fiber optics. [3 marks] (b) (i) What is the condition to be fulfilled by the **refractive indices of core and cladding** of the fiber optic cable for successful data transmission. [2 marks] (ii) Briefly explain why **optical Bandwidth is larger** than that of the electrical Bandwidth of same capacity. [3 marks] (iii) State **3 losses** in fiber optics. [3 marks] (c) Standard size of step index fiber in cable based is given as: Diameter of core $= 50\mu\mathrm{m}$; Diameter of cladding $= 125\mu\mathrm{m}$. The typical value of core and cladding refractive indexes are 1.5 and 1.49 respectively. Determine: (I) **Numerical Aperture (NA)** (II) **Fractional difference in the refractive index**. (III) The **velocity of light** in the optically active region of the material. [5 marks]

Question 6

(a) (i) State the **three mechanisms** used to transmit radio waves from transmitting antenna to receiving antenna. [3 marks] (ii) State the types of information that can be obtained through the **pointing vector**. [2 marks] (b) The electric field of a travelling electromagnetic wave is given by $E(z,t) = 10 \times \cos(\pi \times 10^7 t + \pi z/15); \mathbf{a}_x$ and the intrinsic impedance of the medium is 98.7Ω. Determine: (i) The **direction of wave propagation** (ii) the **wave frequency**, f (iii) The **phase change coefficient**, β (iv) its **wavelength**, λ (v) Its **phase velocity**, v_P (vi) **Ratio between relative permittivity and permeability** of the medium. (vii) The **time average power** carried by the wave. [8 marks] (c) (i) Briefly explain the **operation of a simple electromagnetic relay** in relation to energy transfer in this system. [3 marks] (ii) Write the **energy balance equation** for the motor action. [2 marks] (d) (i) State the **main advantage of electromechanical energy**. [2 marks] (ii) Why do you think that the **heat energy** generated by either motor action or generator action is **irreversible conversion**? [2 marks]